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1**Physiology and levels of function**

COMP

You should be able to: Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

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2**Structure and function relationship**

COMP

You should be able to: Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

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3**Body fluid compartments**

COMP

You should be able to: State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments.

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4**Compartment separation and clinical volume shifts**

COMP

You should be able to: Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.

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5**Homeostasis defined**

COMP

You should be able to: Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

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6**Mass balance**

COMP

You should be able to: Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

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7**Mass flow and clearance**

COMP

You should be able to: Calculate mass flow as concentration times volume flow, and clearance as rate of removal divided by plasma concentration, and explain why two patients with the same concentration can have very different delivery.

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8**Feedback loop components**

COMP

You should be able to: Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

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9**Local and reflex control pathways**

COMP

You should be able to: Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

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10**Negative and positive feedback**

COMP

You should be able to: Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

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11**Feedforward control and acclimatization**

COMP

You should be able to: Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

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12**Signal types and range**

COMP

You should be able to: Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

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13**Receptor location and ligand solubility**

COMP

You should be able to: Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

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14**Signal amplification**

COMP

You should be able to: Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

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15**Receptor modulation**

COMP

You should be able to: Define agonist, antagonist, competitive inhibition, up regulation, and down regulation, and predict the effect of chronic ligand excess on target cell sensitivity.

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16**Experimental design and controls**

COMP

You should be able to: Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

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17**Measurement error and variability**

COMP

You should be able to: Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

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18**Graphing and data interpretation**

COMP

You should be able to: Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

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